📊 Function Decomposition Report

Sales Report Generator Refactoring Exercise

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1. Introduction

This report documents the refactoring of a complex Python function (generate_sales_report) into smaller, modular helper functions. The original function was responsible for multiple tasks including validation, filtering, aggregation, grouping, and report generation. The goal of this exercise was to apply function decomposition principles to improve readability, maintainability, and testability without changing the external behaviour of the system.

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2. Original Problem

The original generate_sales_report function was a monolithic implementation performing several responsibilities in a single block of code:

- Input validation
- Date range filtering
- Data filtering based on conditions
- Metric calculations (total, average, min, max)
- Data grouping by category/product/region/customer
- Report formatting (summary, detailed, forecast)
- Chart data generation
- Output formatting (JSON, HTML, Excel, PDF)

Issues Identified:

- High complexity and long function length
- Difficult to read and understand
- Hard to debug and maintain
- Low reusability of logic
- Difficult to test individual components

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3. Refactoring Approach

The function was decomposed using the **Single Responsibility Principle (SRP)**. Each logical operation was extracted into a dedicated helper function.

Steps followed:

1. Identified distinct responsibilities in the original function
2. Grouped related logic into functional blocks
3. Extracted reusable helper functions
4. Simplified the main function to act as an orchestrator
5. Ensured behaviour remained unchanged

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4. Decomposed Functions

4.1 Input Validation

Responsible for validating input parameters.

```
def validate_inputs(sales_data, report_type, output_format):
    if not sales_data or not isinstance(sales_data, list):
        raise ValueError("Sales data must be a non-empty list")

    if report_type not in ['summary', 'detailed', 'forecast']:
        raise ValueError("Invalid report type")

    if output_format not in ['pdf', 'excel', 'html', 'json']:
        raise ValueError("Invalid output format")
```

4.2 Date Filtering

Filters sales records based on a given date range.

```
def filter_by_date(sales_data, date_range):
    if not date_range:
        return sales_data

    start_date = datetime.strptime(date_range['start'], '%Y-%m-%d')
    end_date = datetime.strptime(date_range['end'], '%Y-%m-%d')

    return [
        sale for sale in sales_data
        if start_date <= datetime.strptime(sale['date'], '%Y-%m-%d') <= end_date
    ]
```

4.3 Additional Filtering

Applies dynamic filters such as category, region, or product.

```
def apply_filters(sales_data, filters):
    if not filters:
        return sales_data

    for key, value in filters.items():
        if isinstance(value, list):
            sales_data = [s for s in sales_data if s.get(key) in value]
        else:
            sales_data = [s for s in sales_data if s.get(key) == value]

    return sales_data
```

4.4 Metrics Calculation

Calculates key business metrics.

```
def calculate_metrics(sales_data):
    total = sum(s['amount'] for s in sales_data)
    avg = total / len(sales_data)
    max_sale = max(sales_data, key=lambda x: x['amount'])
    min_sale = min(sales_data, key=lambda x: x['amount'])

    return {
        "total": total,
        "average": avg,
        "max": max_sale,
        "min": min_sale
    }
```

4.5 Data Grouping

Groups sales data by a selected field.

```
def group_sales(sales_data, grouping):
    grouped = {}

    if not grouping:
        return grouped
```

```
    for sale in sales_data:
        key = sale.get(grouping, 'Unknown')

        if key not in grouped:
            grouped[key] = {"count": 0, "total": 0}

        grouped[key]["count"] += 1
        grouped[key]["total"] += sale["amount"]

    for key in grouped:
        grouped[key]["average"] = grouped[key]["total"] / grouped[key]["count"]

    return grouped
```

4.6 Report Builder

Constructs the final report structure.

```
def build_summary_report(metrics, sales_data, report_type):
    return {
        "report_type": report_type,
        "summary": {
            "total_sales": metrics["total"],
            "transaction_count": len(sales_data),
            "average_sale": metrics["average"],
            "max_sale": metrics["max"],
            "min_sale": metrics["min"]
        }
    }
```

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5. Refactored Main Function

The main function is now simplified and acts as an orchestrator:

```
def generate_sales_report(
    sales_data,
    report_type='summary',
    date_range=None,
    filters=None,
    grouping=None,
    include_charts=False,
    output_format='json'
):
```

```
    validate_inputs(sales_data, report_type, output_format)

    sales_data = filter_by_date(sales_data, date_range)
    sales_data = apply_filters(sales_data, filters)

    if not sales_data:
        return {"message": "No data matches the specified criteria", "data": []}

    metrics = calculate_metrics(sales_data)
    grouped = group_sales(sales_data, grouping)

    report = build_summary_report(metrics, sales_data, report_type)

    if grouping:
        report["grouping"] = grouped

    return report
```

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6. Benefits of Refactoring

6.1 Improved Readability

The logic is now split into meaningful functions, making the code easier to understand.

6.2 Better Maintainability

Each function handles a single responsibility, making updates easier and safer.

6.3 Reusability

Functions like calculate_metrics and apply_filters can be reused in other analytics systems.

6.4 Easier Testing

Each function can now be tested independently, improving test coverage and debugging efficiency.

6.5 Reduced Complexity

The main function is significantly shorter and acts only as a coordinator.

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7. Reflection

7.1 How did breaking down the function improve readability and maintainability?

Breaking the function into smaller units made the code easier to follow because each function has a single clear responsibility. This separation reduces cognitive load and improves maintainability.

7.2 What was the most challenging part of decomposing the function?

The most challenging part was identifying clear boundaries between responsibilities, especially separating filtering logic from metrics calculation while maintaining the original behaviour.

7.3 Which extracted function is most reusable?

The calculate_metrics() function is the most reusable because it can be applied to any dataset requiring basic statistical analysis, not only sales data.

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8. Conclusion

The refactoring exercise successfully transformed a complex monolithic function into a modular and maintainable system. The improved structure enhances readability, reusability, and scalability while preserving the original functionality and passing all existing tests.